Preparation of Tritium Spiked Water

1 Acquire a clean Container

Ensure that the container is clean, sterilized and dry.

1.1 Weigh the empty container

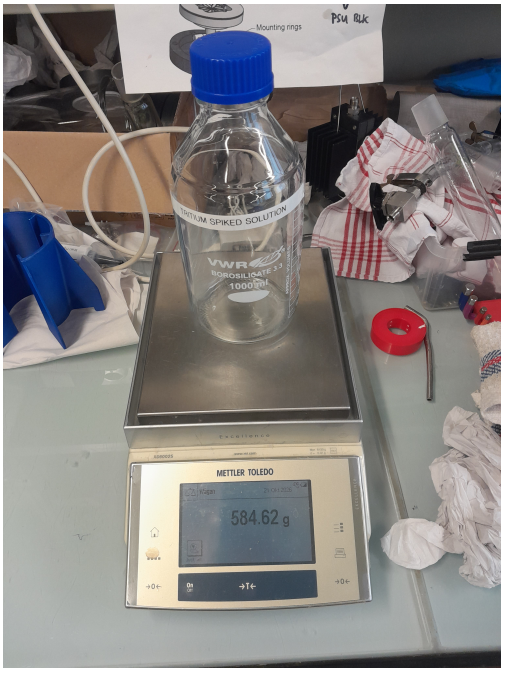


Figure 1: Weighing of the clean empty container

For example the mass of the bottle as displayed in the figure m_b= 584.62 g.

1.2 Add tritium free water

Tare the balance and add a target amount of tritium free water to the bottle and weigh it.

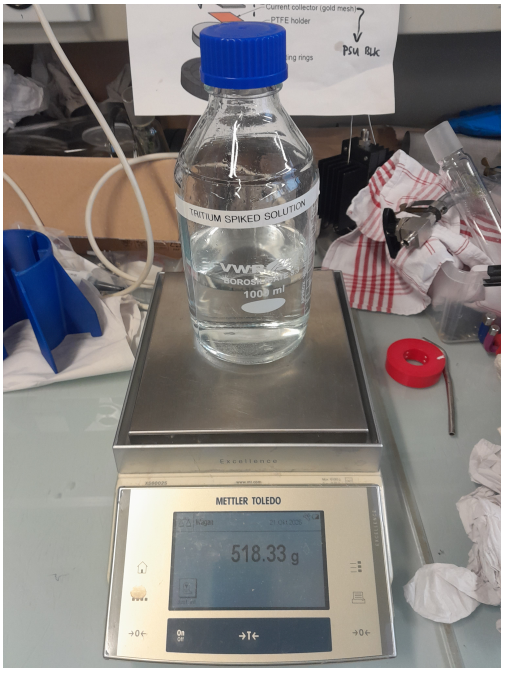


Figure 2: Weighing of tritium free water

For example the mass of tritium free water as displayed in the figure m_{tfw}= 518.33 g

1.3 Add the tritium standard

Tare the balance and add a target amount of the tritium standard to the bottle and weigh it.

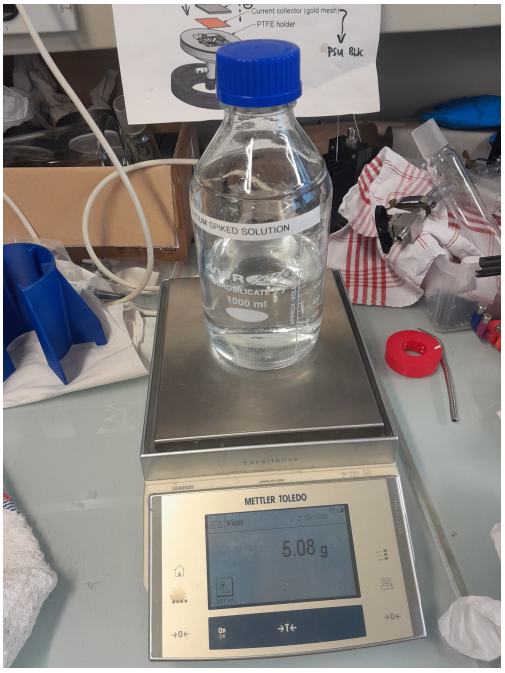


Figure 3: Weighing of tritium standard

For example the mass of tritium standard as displayed in the figure m_s= 5.08 g

1.4 Compute the final activity

- Use the decay equation to determine the current activity of the standard.

$$T(t) = T_0 \left(\frac{1}{2}\right)^{t/t_{1/2}}, \tag{1}$$

  take $t_{1/2}$= 4500 days

- Use the concentration relation to determine the final activity of the spiked solution

$$T_f = \frac{T_s m_s}{m_s + m_{tfw}} \tag{2}$$

  where T_f is the final activity and T_s is the activity of the standard spiked solution, T_s is the current activity of the standard, m_s is the mass of the standard and m_{tfw} is the mass of tritium free water.

- The 15046 TU (REF 24.10.2018) standard was used. Therefore as of 21/10/2026 the activity of the spiked solution in figure 3 is 93.15 TU.